

DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II
WEEK 10: LECTURE NOTE

CHAPTER 4. INTEGRATION

Antiderivatives and Indefinite Integration

Definition 1. A function F is **an antiderivative** of f on an interval I when

$$F'(x) = f(x) \quad \text{for all } x \in I.$$

Remark 2. Note that F is called **an antiderivative** of f , rather than **the antiderivative** of f . To see why, observe that

$$F_1(x) = x^3, \quad F_2(x) = x^3 - 5, \quad \text{and} \quad F_3(x) = x^3 + 97$$

are all antiderivatives of $f(x) = 3x^2$. In fact, for any constant C , the function $F(x) = x^3 + C$ is an antiderivative of f .

Theorem 3. If F is an antiderivative of f on an interval I , then G is an antiderivative of f on the interval I if and only if G is of the form $G(x) = F(x) + C$, for all x in I where C is a constant.

You can represent the entire family of antiderivatives of a function by adding a constant to a known antiderivative. For example, knowing that

$$\frac{d}{dx}[x^2] = 2x$$

you can represent the family of all antiderivatives of $f(x) = 2x$ by

$$G(x) = x^2 + C$$

where C is called **constant of integration**. The family of functions represented by G is the **general solution of the differential equation**

$$G'(x) = 2x$$

A **differential equation** in x and y is an equation that involves x, y , and derivatives of y . For instance,

$$y' = 3x, \quad y' = x^2 + 1, \quad \text{and} \quad y'' + 3y' + 10y = 4x$$

are examples of differential equations.

Example 4. Find the general solution of the differential equation $y' = 2$.

Solution. To begin, we need to find a function whose derivative is 2. One such function is

$$y = 2x.$$

Now, we can use the Theorem above to conclude that the general solution of the differential equation is

$$y = 2x + C.$$

When solving a differential equation of the form

$$\frac{dy}{dx} = f(x)$$

it is convenient to write it in the equivalent differential form

$$dy = f(x)dx.$$

The operations of finding all solutions of this equation is called **antidifferentiation** (or **indefinite integration**) and is denoted by an integral sign $\int$. The general solution is

$$y = \int f(x)dx = F(x) + C.$$

The expression $\int f(x)dx$ is read as **the antiderivative** of f with respect to x . So, the differential dx serves to identify x as the variable of integration. The term **indefinite integral** is a synonym for **antiderivative**.

The inverse nature of integration and differentiation can be verified by substituting $F'(x)$ for $f(x)$ in the indefinite integration to obtain

$$\int F'(x)dx = F(x) + C.$$

Moreover, if $\int f(x)dx = F(x) + C$, then

$$\frac{d}{dx} \left[\int f(x)dx \right] = f(x).$$

These two equations allow you to obtain integration formulas directly from differentiation formulas, as shown in the following summary.

BASIC INTEGRATION RULES

Differentiation Formula

- (1) $\frac{d}{dx} [c] = 0$
- (2) $\frac{d}{dx} [kx] = k$
- (3) $\frac{d}{dx} [kf(x)] = kf'(x), \quad k \in \mathbb{R}$
- (4) $\frac{d}{dx} [f(x) \pm g(x)] = f'(x) \pm g'(x)$
- (5) $\frac{d}{dx} [x^n] = nx^{n-1}$
- (6) $\frac{d}{dx} [e^{ax}] = ae^{ax}$
- (7) $\frac{d}{dx} [e^x] = e^x$
- (8) $\frac{d}{dx} [\sin x] = \cos x$

Integration Formula

- $\int 0dx = c$
- $\int kdx = kx + C$
- $\int kf'(x)dx = kf(x) + C$
- $\int [f(x) \pm g(x)]dx = \int f(x)dx \pm \int g(x)dx$
- $\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$
- $\int e^{ax} dx = \frac{1}{a}e^{ax} + C$
- $\int e^x dx = e^x + C$
- $\int \cos x dx = \sin x + C$

$$\begin{array}{ll}
(9) \quad \frac{d}{dx} [\cos x] = -\sin x & \int \sin x dx = -\cos x + C \\
(10) \quad \frac{d}{dx} [\tan x] = \sec^2 x & \int \sec^2 x dx = \tan x + C \\
(11) \quad \frac{d}{dx} [\sec x] = \sec x \tan x & \int \sec x \tan x dx = \sec x + C \\
(12) \quad \frac{d}{dx} [\cot x] = -\csc^2 x & \int \csc^2 x dx = -\cot x + C \\
(13) \quad \frac{d}{dx} [\csc x] = -\csc x \cot x & \int \csc x \cot x dx = -\csc x + C
\end{array}$$

Example 5 (Describing Antiderivatives).

$$\begin{array}{ll}
\int 3x dx = 3 \int x dx & \text{Constant Multiple Rule} \\
= 3 \int x^1 dx & \text{Rewrite as } x^1 \\
= 3 \frac{x^2}{2} + C & \text{Power Rule} \\
= \frac{3}{2} x^2 + C & \text{Simplify.}
\end{array}$$

The antiderivatives of $3x$ are of the form $\frac{3}{2}x^2 + C$, where C is any constant.

Remark 6. When indefinite integrals are evaluated, a strict application of the basic integration rules tends to produce complicated constants of integration. For instance, in the above Example, the solution could have been written as

$$\int 3x dx = 3 \int x dx = 3\left(\frac{x^2}{2} + C\right) = \frac{3}{2}x^2 + 3C.$$

Because C represents any constant, it is both cumbersome and unnecessary to write $3C$ as the constant of integration. So, $\frac{3}{2}x^2 + 3C$ is written in the simpler form $\frac{3}{2}x^2 + C$.

Example 7. Rewriting Before Integrating

Original Integral	Rewrite	Integrate	Simplify
$\int \frac{1}{x^3} dx$	$\int x^{-3} dx$	$\frac{x^{-2}}{-2} + C$	$-\frac{1}{2x^2} + C$
$\int \sqrt{x} dx$	$\int x^{\frac{1}{2}} dx$	$\frac{x^{\frac{3}{2}}}{\frac{3}{2}} + C$	$\frac{2}{3}x^{\frac{3}{2}} + C$
$\int 2 \sin x dx$	$2 \int \sin x dx$	$2(-\cos x) + C$	$-2 \cos x + C$

Example 8. Integrating Polynomial Functions

$$\begin{array}{ll}
(a) \quad \int dx = \int 1 dx = x + C \\
(b) \quad \int (x + 2) dx = \int x dx + \int 2 dx = \frac{x^2}{2} + C_1 + 2x + C_2 = \frac{x^2}{2} + 2x + C \\
(c) \quad \int (3x^4 - 5x^2 + x) dx = 3 \left(\frac{x^5}{5} \right) - 5 \left(\frac{x^3}{3} \right) + \frac{x^2}{2} + C = \frac{3}{5}x^5 - \frac{5}{3}x^3 + \frac{1}{2}x^2 + C
\end{array}$$

Example 9. Rewriting Before Integrating

$$\begin{aligned}
\int \frac{x+1}{\sqrt{x}} dx &= \int \frac{x}{\sqrt{x}} dx + \int \frac{1}{\sqrt{x}} dx \\
&= \int x^{\frac{1}{2}} dx + \int x^{-\frac{1}{2}} dx \\
&= \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{x^{\frac{2}{2}}}{\frac{2}{2}} + C \\
&= \frac{2}{3} x^{\frac{3}{2}} + 2x^{\frac{3}{2}} + C \\
&= \frac{2}{3} \sqrt{x}(x+3) + C
\end{aligned}$$

Rewrite as two fractions.

Rewrite with fractional exponents.

Integrate.

Simplify

WARNING! When integrating quotients, do not integrate the numerator and denominator separately. This is no more valid in integration than it is in differentiation.

Example 10. Rewriting Before Integrating

$$\begin{aligned}
\int \frac{\sin x}{\cos^2 x} dx &= \int \left(\frac{1}{\cos x} \right) \left(\frac{\sin x}{\cos x} \right) dx \\
&= \int \sec x \tan x dx \\
&= \sec x + C
\end{aligned}$$

Rewrite as a product.

Rewrite using trigonometric identities.

Integrate.

Initial Conditions and Particular Solutions

You have already seen that the equation $y = \int f(x) dx$ has many solutions (each differing from the others by a constant). This means that the graphs of any two antiderivatives of f are vertical translations of each other. For example,

$$y = \int (3x^2 - 1) dx = x^3 + x + C \quad \text{General solution.}$$

Each of these antiderivatives is a solution of the differential equation

$$\frac{dy}{dx} = 3x^2 - 1.$$

In many applications of integration, you are given enough information to determine a **particular solution**. To do this, you need only know the value of $y = F(x)$ for one value of x . This information is called an **initial condition**. Only one curve passes through the point $(2, 4)$. To find this curve, you can use the general solution

$$F(x) = x^3 - x + C \quad \text{General solution}$$

and the initial condition

$$F(2) = 4 \quad \text{Initial condition.}$$

By using the initial condition in the general solution, you can determine that

$$F(2) = 2^3 - 2 + C = 4$$

which implies that $C = -2$. So, you obtain

$$F(x) = x^3 - x - 2 \quad \text{Particular solution}$$

Example 11. Find the general solution of

$$F'(x) = \frac{1}{x^2}, \quad x > 0$$

and find the particular solution that satisfies the initial condition $F(1) = 0$.

Solution. In class

INTEGRATION continued.....

The Fundamental Theorem of Calculus

Theorem 12 (The Fundamental Theorem of Calculus). If a function f is continuous on the closed interval $[a, b]$ and F is an antiderivative of f on the interval $[a, b]$, then the definite integral

$$\int_a^b f(x)dx = F(b) - F(a).$$

Remark 13. Guidelines for using the Fundamental Theorem of Calculus

1. Provided you can find an antiderivative of f , you now have a way to evaluate a **definite integral**.
2. When applying the Fundamental Theorem of Calculus, the notation shown below is convenient.

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a).$$

3. It is not necessary to include a constant of integration in the antiderivative.

$$\int_a^b f(x)dx = [F(x) + C]_a^b = [F(b) + C] - [F(a) + C] = F(b) - F(a).$$

Example 14. Evaluate each definite integral.

1. $\int_1^2 (x^2 - 3)dx$

2. $\int_1^4 3\sqrt{x}dx$

3. $\int_0^{\frac{\pi}{4}} \sec^2 x \, dx$

Example 15 (A Definite Integral Involving Absolute Value). Evaluate the definite integral

$$\int_0^2 |2x - 1|dx.$$

Solution. In class.

Theorem 16 (The Second Fundamental Theorem of Calculus). If f is continuous on an open interval I containing a , then, for every x in the interval,

$$\frac{d}{dx} \left[\int_a^x f(t)dt \right] = f(x).$$

Example 17. Find the derivative of $F(x) = \int_0^x 3t^2 dt$.

Solution. In class.

Example 18. Use *The Second Fundamental Theorem of Calculus* to evaluate the following derivatives.

1. $\frac{d}{dx}[\int_0^x \sqrt{t^2 + 1} dt] = \sqrt{x^2 + 1}$

2. $\frac{d}{dx}[\int_{\frac{\pi}{2}}^x \cos t dt] = \cos x$

3. $\frac{d}{dx}[\int_{-1}^x t dt] = x$

Example 19. Use *The Second Fundamental Theorem of Calculus* to evaluate

$$\frac{d}{dx}[\int_0^{x^2} 5t dt].$$

Solution. In class

Integration by Substitution

Theorem 20 (Antidifferentiation of a Composite Function). Let g be a function whose range is an interval I , and let f be a function that is continuous on I . If g is differentiable on its domain and F is an antiderivative of f on I , then

$$\int f(g(x))g'(x)dx = F(g(x)) + C.$$

Letting $u = g(x)$ gives $du = g'(x)dx$, and

$$\int f(u)du = F(u) + C.$$

Example 21. Find $\int (x^2 + 1)^2(2x)dx$.

Solution. In class.

Example 22. Find the following indefinite integrals.

(a) $\int 5 \cos 5x dx$

(b) $\int x(x^2 + 1)^2 dx$

(c) $\int \sqrt{2x - 1} dx$

GUIDELINES FOR MAKING A CHANGE OF VARIABLES

1. Choose a substitution $u = g(x)$. Usually, it is best to choose the inner part of a composite function, such as a quantity raised to a power.
2. Compute $du = g'(x)dx$.

3. Rewrite the integral in terms of the variable u .
4. Find the resulting integral in terms of u .
5. Replace u by $g(x)$ to obtain an antiderivative in terms of x .
6. Check your answers by differentiating.

Example 23. Find $\int x\sqrt{2x-1}dx$.

Solution. In class.

Example 24. Find $\int \sin^2 3x \cos 3x dx$.

Solution. In class.

Theorem 25. The General Power Rule for Integration

If g is a differentiable function of x , then

$$\int [g(x)]^n g'(x) dx = \frac{1}{n+1} [g(x)]^{n+1} + C, \quad n \neq -1.$$

Equivalently, if $u = g(x)$, then

$$\int u^n du = \frac{u^{n+1}}{n+1} + C, \quad n \neq -1.$$

Example 26. Substitution and the General Power Rule

1. $\int 3(3x-1)^4 dx$
2. $\int (2x+1)(x^2+x) dx$
3. $\int 3x^2 \sqrt{x^3-2} dx$
4. $\int \frac{-4x}{(1-2x^2)^2} dx$
5. $\int \cos^2 x \sin x dx$

Theorem 27. If the function $u = g(x)$ has a continuous derivative on the closed interval $[a, b]$, and f is continuous on the range of g , then

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

Example 28. Change of Variables

Evaluate $\int_0^1 x(x^2+1)^3 dx$.

Solution. In class

Example 29. Change of Variables

Evaluate $\int_1^5 \frac{x}{\sqrt{2x-1}} dx$.

Solution. In class

Theorem 30. Integration of Even and Odd Functions Let f be integrable on the closed interval $[-a, a]$ for a positive number a .

1. If f is an even function, then $\int_{-a}^a f(x)dx = 2 \int_0^a f(x)dx$.

2. If f is an odd function, then $\int_{-a}^a f(x)dx = 0$.

Example 31. Integration of Even and Odd Functions

1. $\int_{-2}^2 3x^2 dx = 2 \int_0^2 3x^2 dx = 2[x^3]_0^2 = 2[2^3 - 0^3] = 16$.

2. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [\sin^3 x \cos x + \sin x \cos x] dx = 0$ because the function

$$f(x) = \sin^3 x \cos x + \sin x \cos x$$

is odd.

Integrals of Trigonometric Functions

Recall the six trigonometric integration rules that correspond directly to differentiation rules:

$$\int \cos x \, dx = \sin x + C \quad (1)$$

$$\int \sin x \, dx = -\cos x + C \quad (2)$$

$$\int \sec^2 x \, dx = \tan x + C \quad (3)$$

$$\int \csc^2 x \, dx = -\cot x + C \quad (4)$$

$$\int \sec x \tan x \, dx = \sec x + C \quad (5)$$

$$\int \csc x \cot x \, dx = -\csc x + C \quad (6)$$

With the Log Rule, we can now complete the set of basic trigonometric integration formulas by using trigonometric identities.

1. $\int \tan x \, dx$

$$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx = - \int \overbrace{\frac{1}{\cos x}}^{\frac{1}{u}} \overbrace{(-\sin x) dx}^{du} = - \int \frac{1}{u} du = -\ln |u| + C = -\ln |\cos x| + C$$

2. $\int \cot x \, dx$

$$\int \cot x \, dx = \int \frac{\cos x}{\sin x} \, dx = \int \overbrace{\frac{1}{\sin x}}^{\frac{1}{u}} \overbrace{(\cos x) dx}^{du} = \int \frac{1}{u} du = \ln |u| + C = \ln |\sin x| + C$$

$$3. \int \sec x \, dx$$

$$\begin{aligned} \int \sec x \, dx &= \int \sec x \left(\frac{\sec x + \tan x}{\sec x + \tan x} \right) dx = \int \overbrace{\frac{1}{\sec x + \tan x}}^{\frac{1}{u}} \overbrace{(\sec^2 x + \sec x \tan x) dx}^{du} \\ &= \int \frac{1}{u} du = \ln |u| + C \\ &= \ln |\sec x + \tan x| + C \end{aligned}$$

$$4. \int \csc x \, dx$$

$$\begin{aligned} \int \csc x \, dx &= \int \csc x \left(\frac{\csc x + \cot x}{\csc x + \cot x} \right) dx = - \int \overbrace{\frac{1}{\csc x + \cot x}}^{\frac{1}{u}} \overbrace{(-\csc^2 x - \csc x \cot x) dx}^{du} \\ &= - \int \frac{1}{u} du = - \ln |u| + C \\ &= - \ln |\csc x + \cot x| + C \end{aligned}$$

SUMMARY OF INTEGRALS OF THE SIX BASIC TRIGONOMETRIC FUNCTIONS

$$\int \cos x \, dx = \sin x + C \quad (7)$$

$$\int \sin x \, dx = -\cos x + C \quad (8)$$

$$\int \tan x \, dx = -\ln |\cos x| + C \quad (9)$$

$$\int \cot x \, dx = \ln |\sin x| + C \quad (10)$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C \quad (11)$$

$$\int \csc x \, dx = -\ln |\csc x + \cot x| + C \quad (12)$$

Example 32. Evaluate $\int_0^{\frac{\pi}{4}} \sqrt{1 + \tan^2 x} \, dx$.

Solution. By using a trigonometric identity $1 + \tan^2 x = \sec^2 x$, we obtain

$$\int_0^{\frac{\pi}{4}} \sqrt{1 + \tan^2 x} \, dx = \int_0^{\frac{\pi}{4}} \sqrt{\sec^2 x} \, dx = \int_0^{\frac{\pi}{4}} \sec x \, dx = [\ln |\sec x + \tan x|]_0^{\frac{\pi}{4}} = \ln(\sqrt{2} + 1) - \ln 1 \approx 0.881$$

Example 33. Finding an Indefinite Integral by u -Substitution

$$1. \text{ Evaluate } \int \tan 5x \, dx$$

$$2. \text{ Evaluate } \int \frac{\sec \theta \tan \theta}{\sec \theta - 1} d\theta.$$

$$3. \text{ Evaluate } \int [\sec 2x + \tan 2x] dx$$